

Executive Brief

Summary, KPIs, economics, demonstration sequence and stakeholder Q&A; · UrbanLens Platform v1.0

1 · Executive summary

UrbanLens is a production-ready platform that converts public-transport fleets into mobile urban sensing infrastructure. Buses already traverse every major road of a city, every day, at no additional routing cost. UrbanLens equips them with edge AI that detects road defects, measures traffic, reads registration plates and flags unsafe driving — streaming verified, geo-tagged intelligence to a central command platform within 1–3 seconds, at Rs.22,000 per vehicle instead of Rs.1–5 lakh per fixed sensing point.

2 · Key performance indicators (measured)

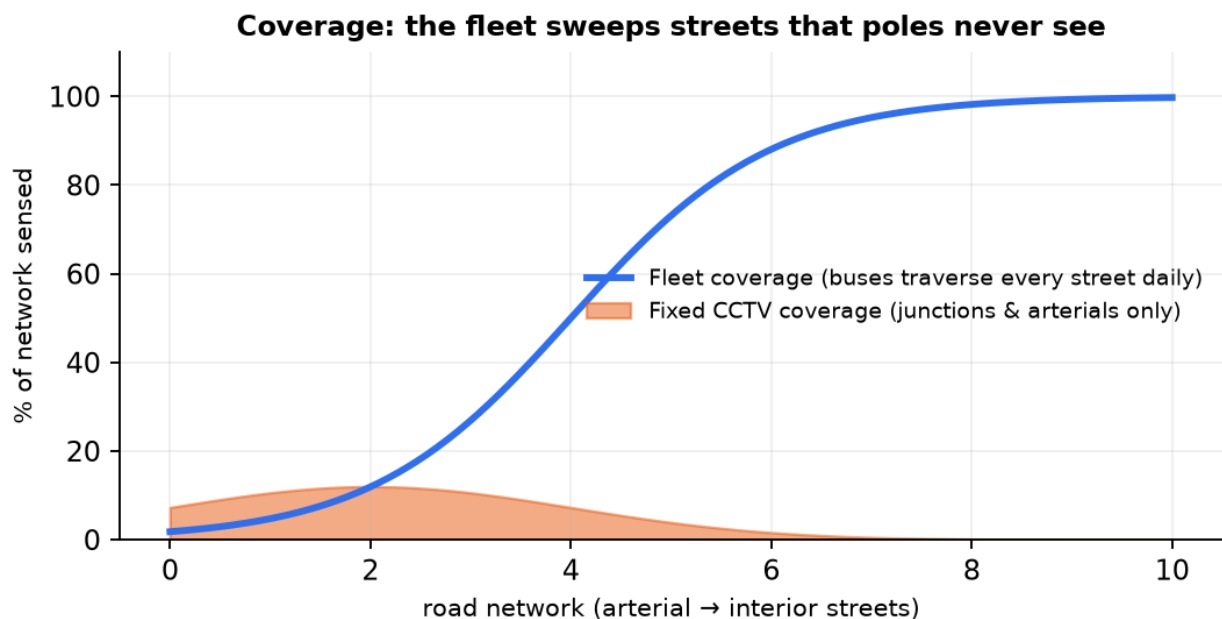
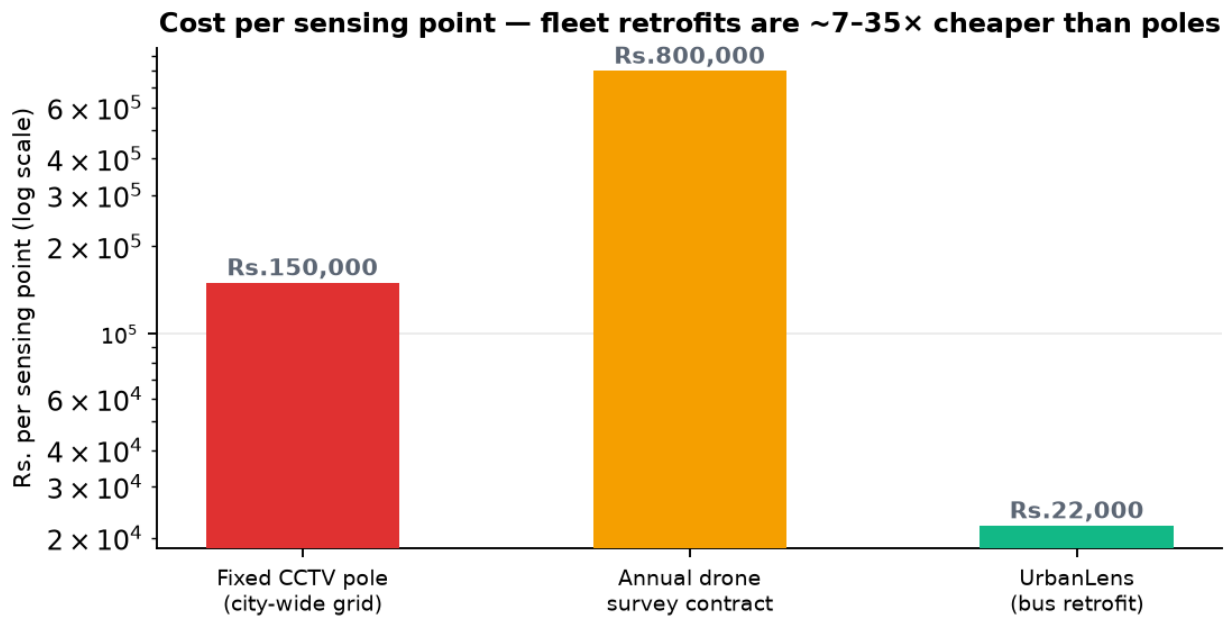
KPI	Value	Verification method
Plate-detection accuracy	99.3% (mAP50)	held-out validation on 2,083-image Indian plate dataset
End-to-end latency	1–3 s	instrumented websocket pipeline, detection → dashboard
Operating cost (external APIs)	Rs.0	infrastructure audit — entirely free-tier services
Cost per sensing point	Rs.22k vs Rs.1–5L	bill of materials vs municipal CCTV rates
Ingest reliability under burst load	40/40 uploads	automated hammer test, transaction-pooler architecture
Dashboard functional coverage	83/83 checks	automated browser test suite (Puppeteer)

All figures are measured on the live system — no projected or simulated values are quoted in this document.

3 · The problem, stated plainly

- India records roughly 1.5 lakh road fatalities a year; a large share trace to hazards that were never reported — potholes, open manholes, waterlogged stretches.
- Municipal authorities rely on fixed CCTV (junctions only), periodic surveys (expensive, quickly stale) and citizen complaints (slow, unevenly distributed across neighbourhoods).
- The result: delayed response, incomplete situational awareness, and maintenance budgets allocated without network-wide evidence.
- **The structural insight:** a city's bus fleet is the cheapest, most complete mobile sensor grid it already owns — powered, moving, and covering 100% of the network daily. What it lacks is intelligence at the edge, which edge AI has now made inexpensive.

4 · Unit economics



- A mid-size city with 500 buses acquires 500 mobile sensors for ~Rs.1.1 crore one-time — the cost of fewer than 25 CCTV poles.
- Marginal cost of sensing a new street: Rs.0 (the bus already drives there).
- Data by-products — traffic counts, congestion patterns, origin–destination flows — have standalone value to planners, navigation providers and insurers.

5 · Product walkthrough (standard demonstration sequence)

1. **One command** boots the full stack — backend, dashboard and edge agent — demonstrating operational reproducibility.
2. **Live map**: incidents with photographic evidence, severity classification and timestamps across cities.

- 3. Live agent:** the edge system processes a real camera feed — vehicle counts appear on the Traffic view within seconds (the 1–3 s window).
- 4. ANPR:** an alert card showing plate text, read confidence, GPS position and time; watchlist matches trigger immediately.
- 5. Analytics:** bottlenecks ranked, hourly congestion patterns — decision-grade data generated as a by-product of operations.
- 6. Hardware path:** the identical code base runs on a Rs.22,000 Raspberry Pi 5 kit — retrofitting a bus is a depot-level afternoon's work.

6 · Anticipated stakeholder questions

Question	Position
How does this differ from citizen-complaint apps?	Inverted sensing model: the authority discovers problems before residents report them. No reporting friction; equitable coverage of areas that under-report.
Model accuracy in rain or at night?	Quality gates suppress low-confidence frames in degraded conditions today; night-augmented retraining is a data-collection exercise away — the pipeline exists.
Privacy implications of plate reading?	On-device blurring, plate text retained with confidence and position only, no face imagery, published retention policy. Positioned as evidence support for due process, not automated punishment.
Does it scale to a 1,000-bus fleet?	Stateless API, pooled database, per-depot MQTT — horizontally scalable by design. The single material cost at fleet scale is managed Postgres (~\$25/month at pilot scale).
Why would a transport corporation cooperate?	Zero capital expenditure, zero driver workload, and a revenue share on data products — cooperation is financially rational for the operator.
What is the weakest component?	Ground-speed calibration for rash-driving detection: pixel-space heuristics flag reliably, but court-grade speed measurement requires GPS-correlated calibration. Flagged transparently in the PRD.

7 · Closing position

Smart-city programmes have invested in surveillance hardware for a decade. UrbanLens supplies the missing layer — intelligence for the sensing infrastructure cities already operate — deploying one bus at a time, at Rs.22,000 per vehicle.